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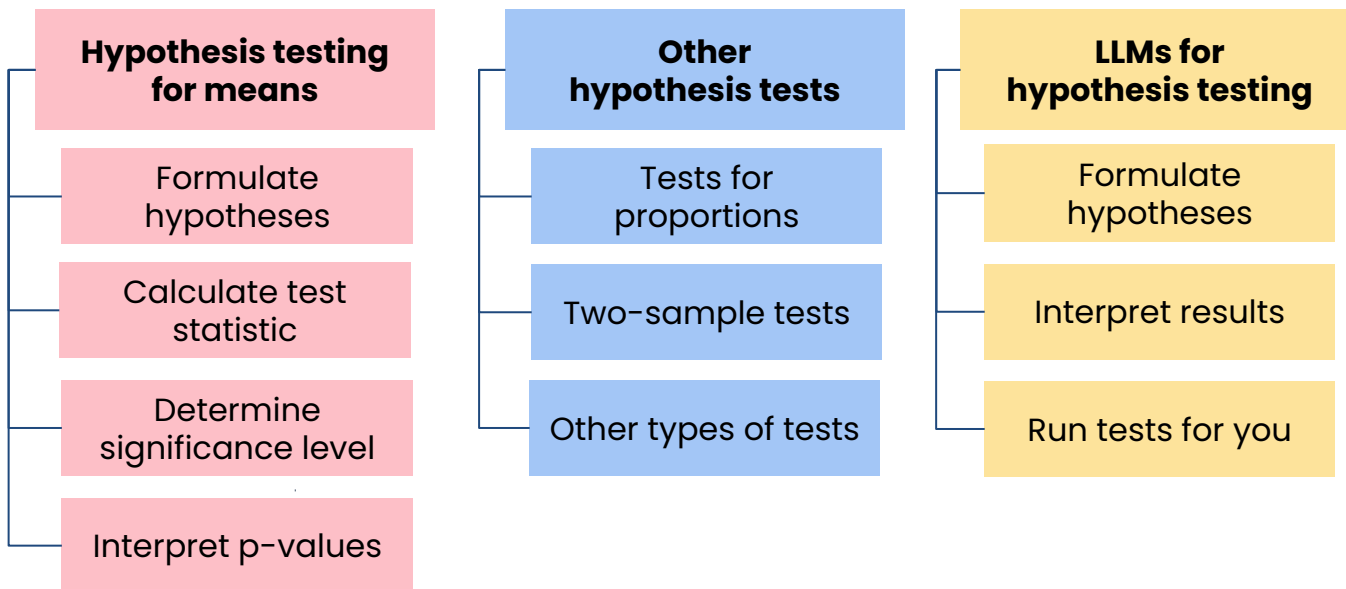
Applied Statistics for Data Analytics

Module 4: Hypothesis testing

Hypothesis testing

Module 4 introduction

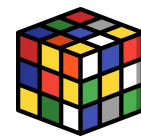
Module 4 outline



Hypothesis testing

Demo: hypothesis testing
in action

Hypothesis testing



Did you observe an outcome because of:

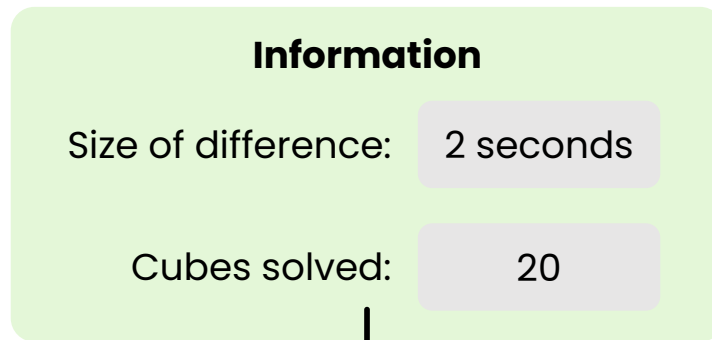


Random chance

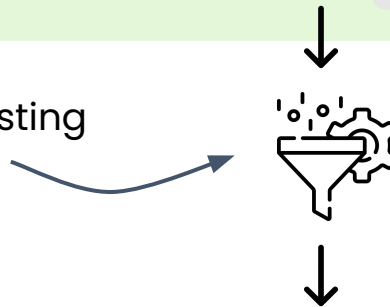
"You got lucky"



True pattern



Hypothesis testing



Who's the better cube solver?



Hypothesis testing

Hypothesis testing: means

Scenario



30 day free trial

Problem: Investigate whether users with a free trial stay subscribed for longer



Users without free trial subscribe for **10 months**



Sample: 100 users who received a free trial

$\bar{x} = 10.4$ months

$s = 2$ months



Close!

Is **0.4 months** large enough to convince you that free trial is effective?

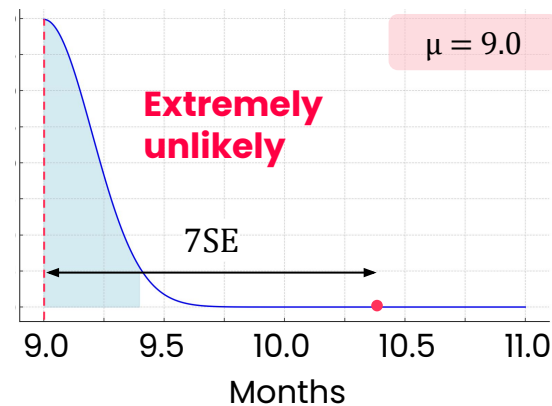
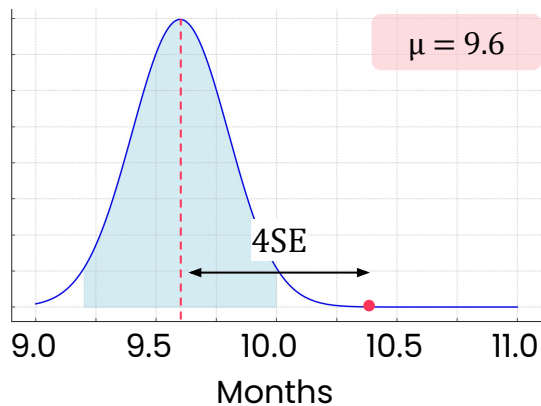
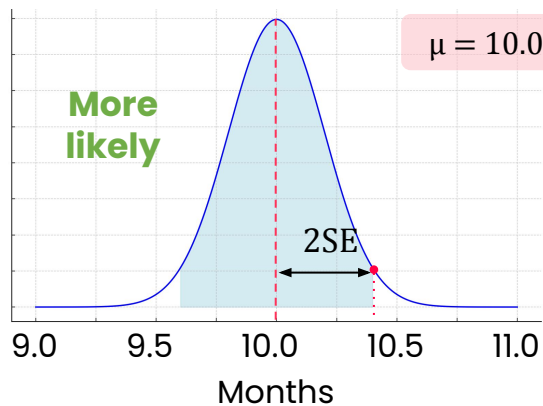
Scenario

$\bar{x} = 10.4$ months

$s = 2$ months

- **Unknown:** Where mean of 10.4 falls in sampling distribution
- **Known:** Sampling distribution for means is normally distributed

$$SE = \frac{s}{\sqrt{n}} = \frac{2}{\sqrt{100}} = 0.2$$



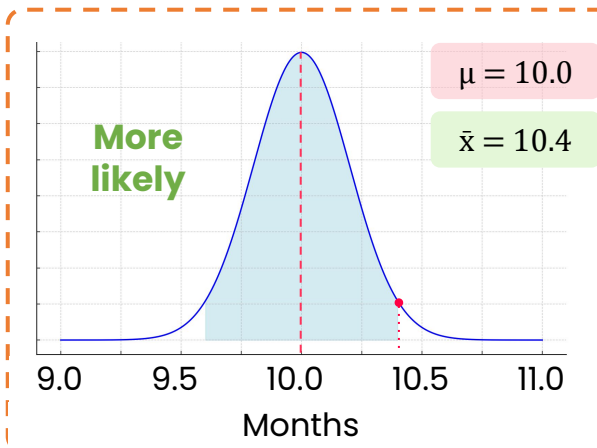
Scenario

❌ **You can never:** know true population mean to compare with

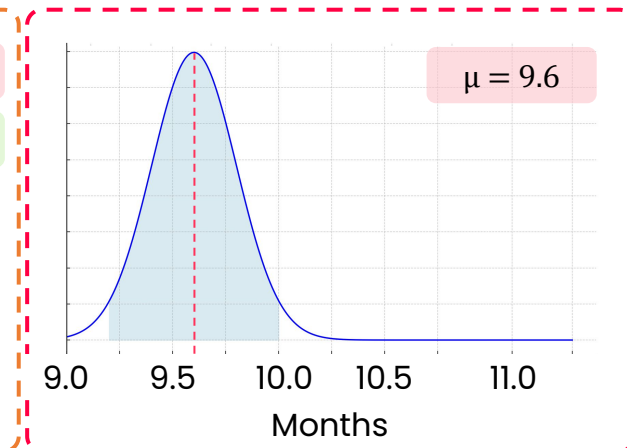
✅ **You can:** calculate likelihood of observing $\bar{x}$ if μ is the value you suspect

Tempting:

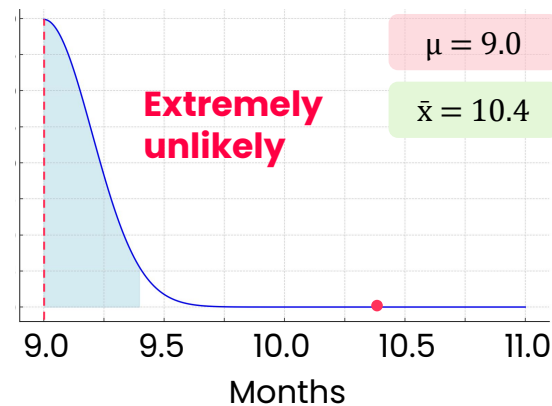
10.4 > 10
↑ ↑
Close!



👎 Cost money, giving no benefit



👎 Slightly worse for retention



Hypothesis testing

- Evaluate whether $\bar{x}$ is significantly different from known μ , given **variability** and **sample size**

Sample mean
for free trial users

$\bar{x} = 10.4$ months

Population mean
for existing users

$\mu = 10.0$ months

- Distinguishes between two possible scenarios:



Observed difference is **due to random chance**



Statistical significance

Not statistically significant

Doesn't help you draw meaningful conclusion



Observed difference **reflects a genuine difference**



Statistically significant

Difference is likely real, can provide evidence for hypothesis

Hypothesis testing

Very common to:



Collect a sample

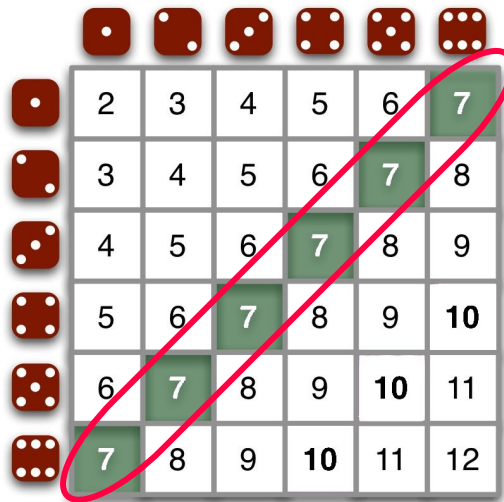


Calculate $\bar{x}$ different from μ



Two values are different: **not enough** to conclude that difference is meaningful

Example: Rolling dice



Population mean:

$$\mu = 7$$

Mean of $n=10$ rolls:

$$\bar{x} = \sim 7$$

Erickson, T. (2011, January 8). An Empirical Approach to Dice Probability. Best Case [Blog post]. Retrieved from <https://bestcase.wordpress.com/2011/01/08/an-empirical-approach-to-dice-probability/>

Hypothesis testing

Only work effectively under certain conditions:

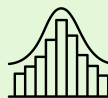
- ✓ Data is a representative sample, ideally a random sample
- ✓ Observations in your data must be independent
- ✓ Must meet one of two conditions:
 - ☐ Normally distributed
 - ☐ Sample size must be large



Sample isn't random: no way of knowing what biases sampling method introduced



"Large": Typically means 30, but 50 or more is ideal



Central Limit Theorem: sampling distribution of the mean approaches normal distribution as n increases

You will learn

How to perform a hypothesis test

- Defining your hypotheses
- Calculating the test statistic
- Defining the significance level
- Calculating the p value
- Interpreting the results



How to define errors



How to work with small sample sizes

Hypothesis testing

The hypothesis

Scenario



30 day free trial

- **Problem:** Mean retention time (μ) of people with free trial
- **Outcomes:**

Hypotheses



$$\mu = 10 \text{ months}$$

Null hypothesis

$$H_0$$

- Aren't able to find evidence that $\mu > 10$
- Finding no effect or no difference

$$\mu > 10 \text{ months}$$

Alternative hypothesis

$$H_1$$

- Evidence that $\mu > 10$
- Free trial was effective

Defining your hypotheses

1. Start with the **null hypothesis**

- Identify value you expect for no effect

No effect of
free trial



$$H_0 : \mu = 10 \text{ months}$$

2. For the **alternative hypothesis**

- Compare population parameter with value in the null hypothesis
- Define the comparison you're interested in



$$H_1 : \mu > 10 \text{ months}$$



Is the mean significantly
greater than 10?

$$H_1 : \mu < 10 \text{ months}$$



Is the mean significantly
less than 10?

$$H_1 : \mu \neq 10 \text{ months}$$



Is the mean significantly
different than 10?

Explaining your hypotheses

Test outcome	Statement	Conclusion
Evidence for H_1	"We reject the null hypothesis"	Null hypothesis is likely not true
No evidence for H_1	"We fail to reject the null hypothesis"	Not enough evidence to reject null hypothesis

You should **avoid** phrases like:

- ❌ "Prove the alternative hypothesis"
- ❌ "Accept the null hypothesis"

This terminology:

- ✅ Helps avoid overstating conclusions
- ✅ Reminds stakeholders that tests can never prove with certainty

Selecting your hypotheses

✓ Hypotheses should be based on:



Theory



Observable evidence

✗ Don't just select them at random

Example:



$$H_0 : \mu = 10 \text{ months}$$

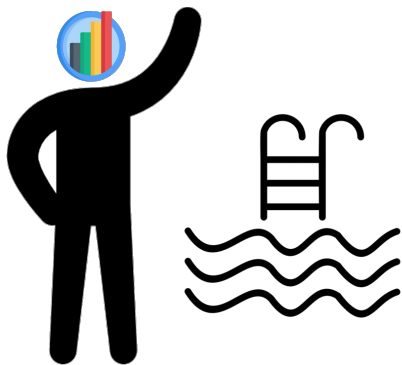
- ✓ Known average for existing users
- ✓ Plausible that the behavior of users who got a free trial is similar

Hypothesis testing

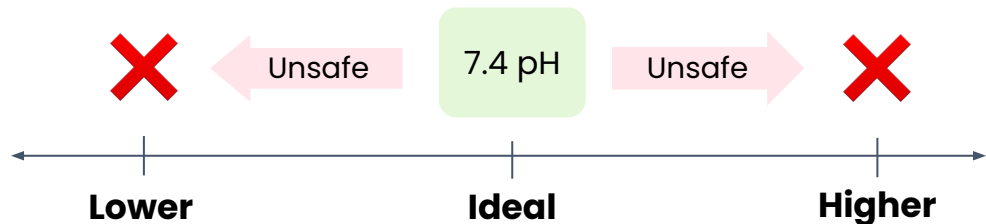
Identifying the hypothesis
and test type

Scenario

Problem: Testing the pH of a swimming pool, for which the ideal pH is 7.4.



You
Data Analyst



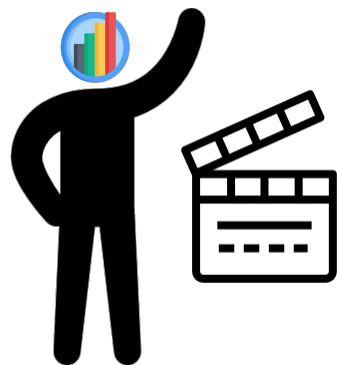
- **Null hypothesis:**

$$H_0 : \mu = 7.4 \text{ pH}$$

- **Alternative hypothesis:**

$$H_1 : \mu \neq 7.4 \text{ pH}$$

Scenario



You
Data Analyst

Problem: Are movie lengths in 2013 greater than 120 mins?

- **Null hypothesis:**

$$H_0 : \mu = 120 \text{ mins}$$

No difference between movies in
2013 & expected value

- **Alternative hypothesis:**

- ☐ Find evidence and reject the null hypothesis
- ☐ Don't find evidence and fail to reject the null hypothesis

$$H_1 : \mu > 120 \text{ mins}$$

Movies are longer
than two hours

Scenario



You
Data Analyst

Problem: Whether average delivery time is less than 45 mins

- **Null hypothesis:**

$$H_0 : \mu = 45 \text{ mins}$$

Delivery times not
different from 45 mins

- **Alternative hypothesis:**

☐ Reject the null hypothesis if
you do find that evidence

☐ Fail to reject it otherwise

$$H_1 : \mu < 45 \text{ mins}$$

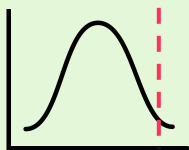
Delivery times are
less than 45 mins

Test types

One-tailed tests

$$\mu > _$$

Reject the H_0 if:

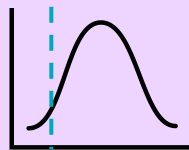


$\bar{x}$ is rare on the upper end

Perform a **right-tailed test**

$$\mu < _$$

Reject the H_0 if:

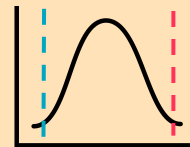


$\bar{x}$ is rare on the lower end

Perform a **left-tailed test**

$$\mu \neq _$$

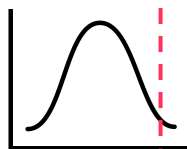
Reject the H_0 if:



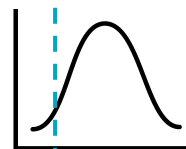
$\bar{x}$ is rare on the either end

Perform a **two-tailed test**

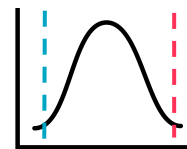
Test types



Right-tailed



Left-tailed



Two-tailed



$$H_1 : \mu \neq 7.4 \text{ pH}$$

Perform a **two-tailed test**



$$H_1 : \mu > 120 \text{ mins}$$

Perform a **right-tailed test**



$$H_1 : \mu < 45 \text{ mins}$$

Perform a **left-tailed test**

Hypothesis testing

Calculating the
test statistic

Scenario

- $n = 50$ movies
- $\bar{x} = 123$ minutes
- $s = 12$ minutes
- $SE = \frac{s}{\sqrt{n}} = \frac{12}{\sqrt{50}} \approx 1.7$

Test statistic

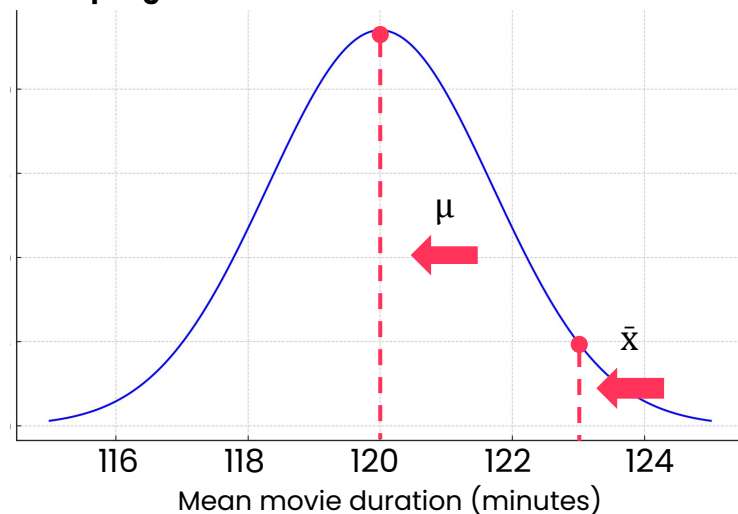
$$z = ?$$

You're performing a **right-tailed** test:

$$H_0 : \mu = 120 \text{ mins}$$

$$H_1 : \mu > 120 \text{ mins}$$

Sampling Distribution of the Mean for Movie Duration



- Highly variable → can't be as confident that means are different
- n is lower → not certain your results really reflect the population

Scenario

- $n = 50$ movies
- $\bar{x} = 123$ minutes
- $s = 12$ minutes

$$\bullet \quad SE = \frac{s}{\sqrt{n}} = \frac{12}{\sqrt{50}} \approx 1.7$$

You're performing a **right-tailed** test:

$$H_0 : \mu = 120 \text{ mins}$$

$$H_1 : \mu > 120 \text{ mins}$$

z-score



Test statistic

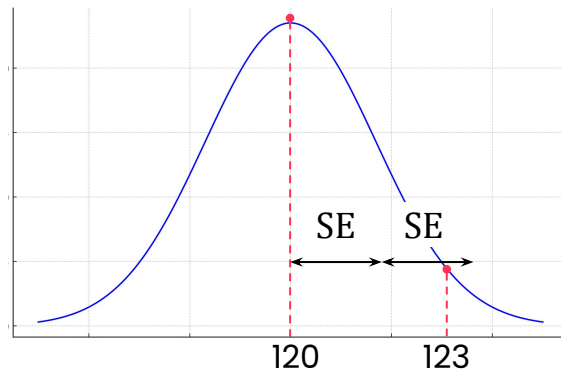
$$z = 1.76$$

1. Calculate difference between $\bar{x}$ and μ

$$\bar{x} - \mu = 123 - 120 = 3$$

2. Divide difference by the standard error

$$= \frac{3}{1.7} = 1.76$$



The test statistic (z)

- Number of σ from μ on the standard normal distribution
- Translating $\bar{x}$ into a value on a standardized scale

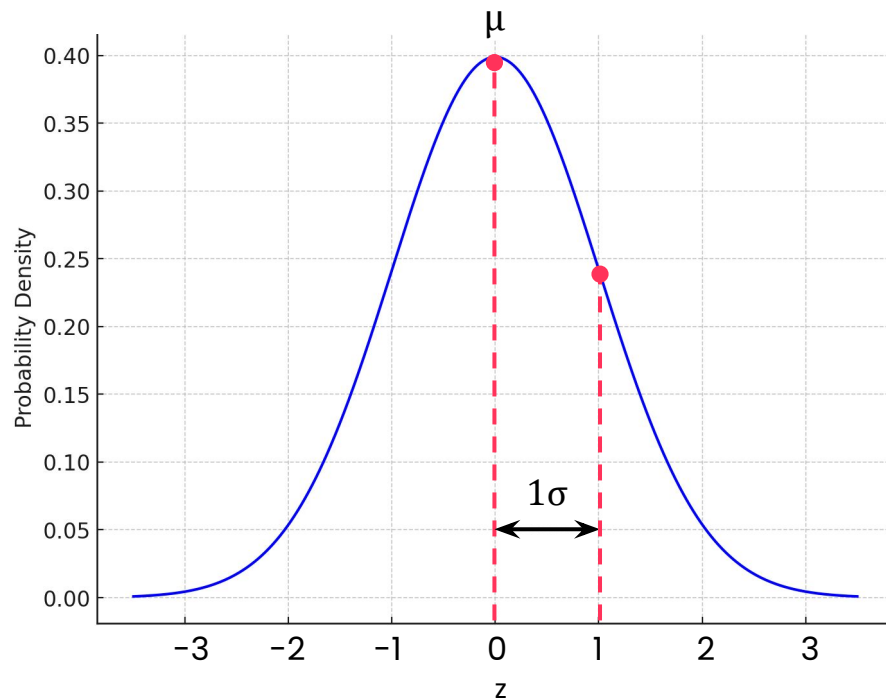
$$\bar{x} = 123$$



$$z = 1.76$$

Where:

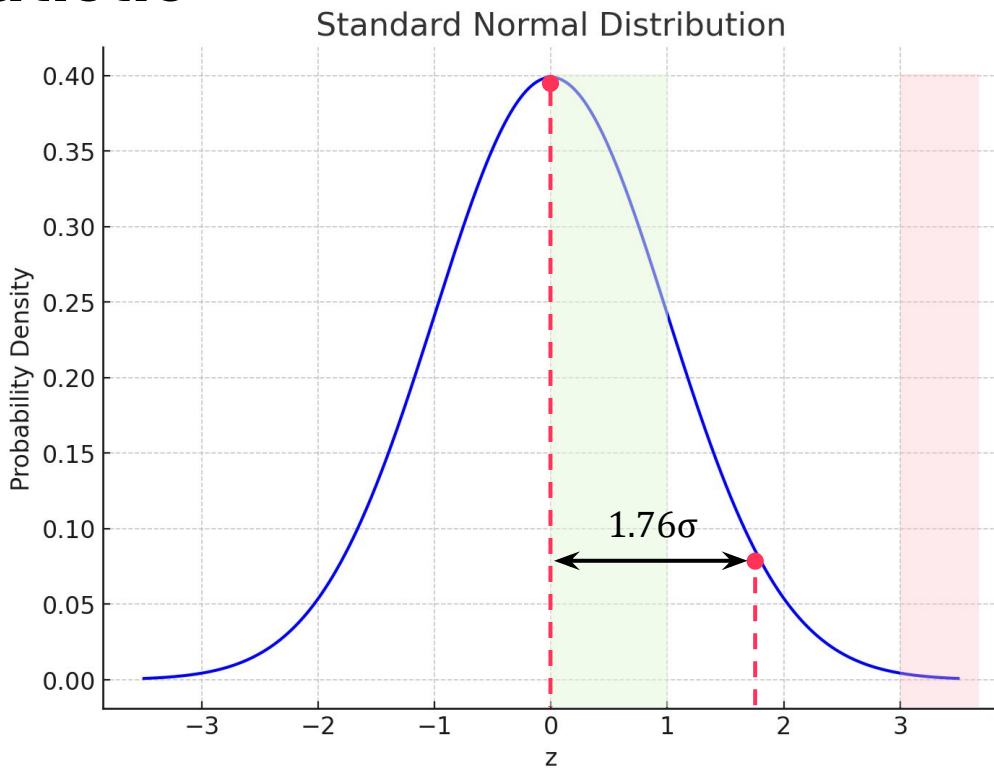
- $\mu = 0$
- Each step is 1σ



Visualizing your test statistic

$$z = 1.76$$

- Not in the **tail** above $z = 3$
- Not in the **middle** set of common values $0 < z < 1$

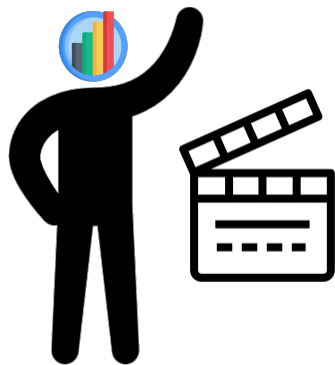


Hypothesis testing

Determining the significance level and rejection region

Scenario

Problem: Test whether movies in 2013 were longer than 120 minutes



You
Data Analyst

Confidence

The level of certainty you have in your conclusions

90%
confidence level

95%
confidence level

99%
confidence level

- Choice depends on how precise your estimate needs to be
- You will have to make this call
- You are trying to **manage** uncertainty



$n = 50$ movies

$$H_0 : \mu = 120 \text{ mins}$$

$\bar{x} = 123$ minutes

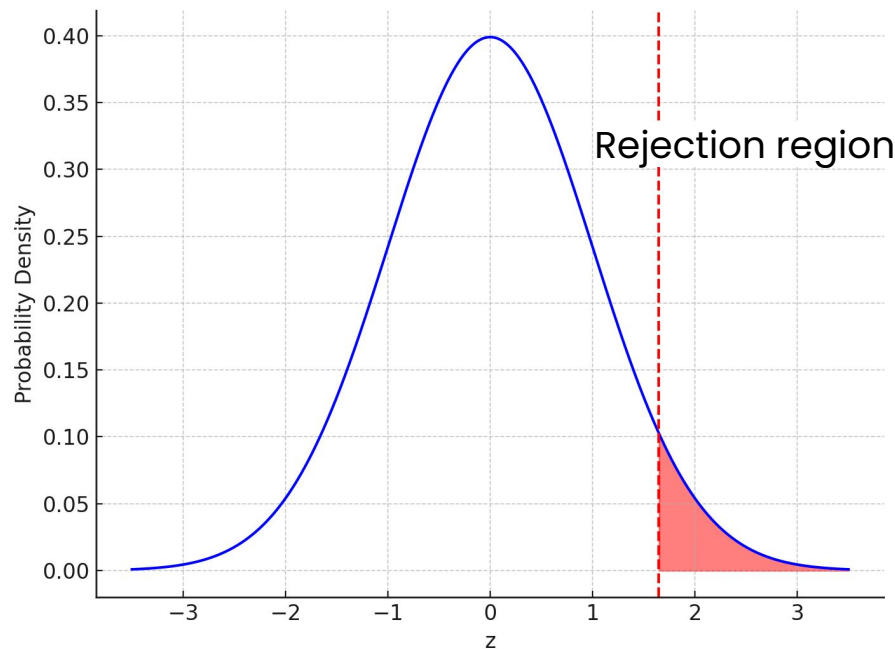
$$H_1 : \mu > 120 \text{ mins}$$

$s = 12$ minutes

$$z = 1.76$$

95% confident that you **correctly** reject H_0 :

- Look for results above the mean that occur **5% of the time or less**
- Any z in this region would lead you to reject H_0



$n = 50$ movies

$$H_0 : \mu = 120 \text{ mins}$$

$\bar{x} = 123$ minutes

$$H_1 : \mu > 120 \text{ mins}$$

$s = 12$ minutes

$$z = 1.76$$

For **one-sided** test with a **95%** confidence:

- Ability to reject H_0 depends on size of rejection region
- $\alpha = 0.05$ is often used in:



Research



Manufacturing

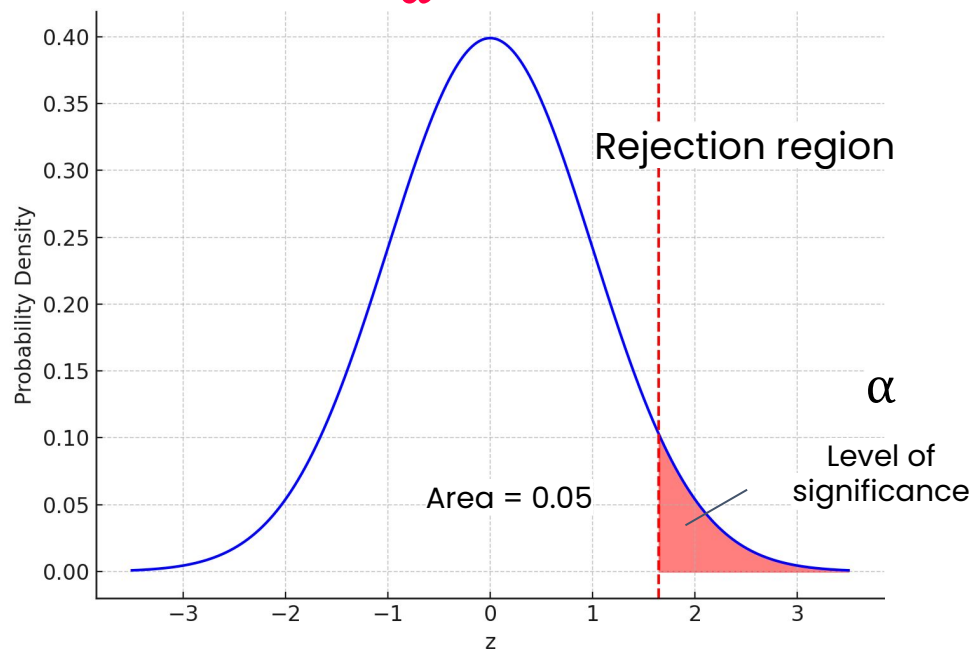


Sciences

Confidence

$$1 - 0.05 = 0.95$$

α





Problem: Be absolutely certain that movies are above 120 minutes before adjusting the schedule

- For a more precise test:

$$\alpha = 0.01$$

99%
confidence level

- Reject H_0 if test statistic is in top 1%



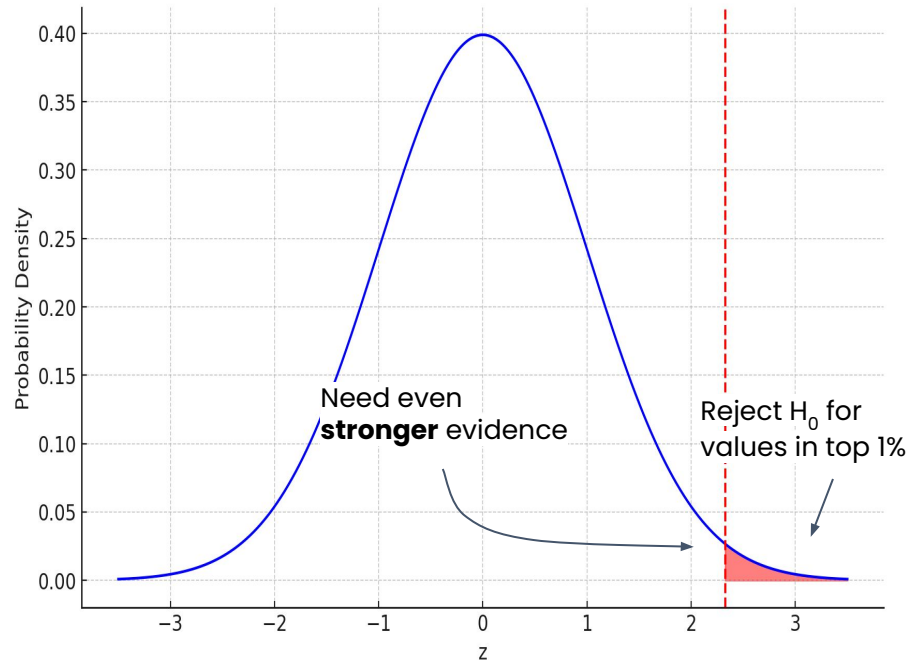
Clinical trials



Impact studies

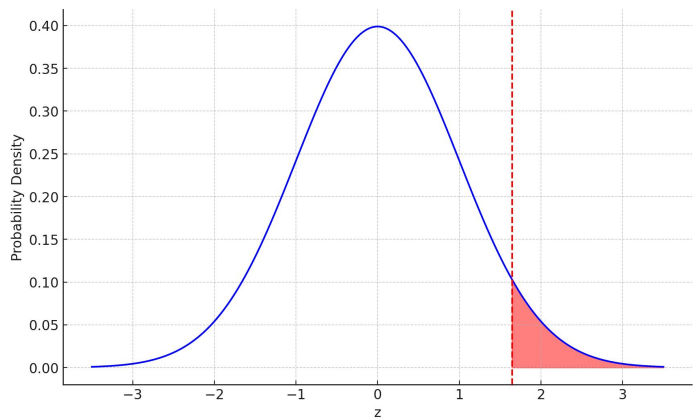


Audits

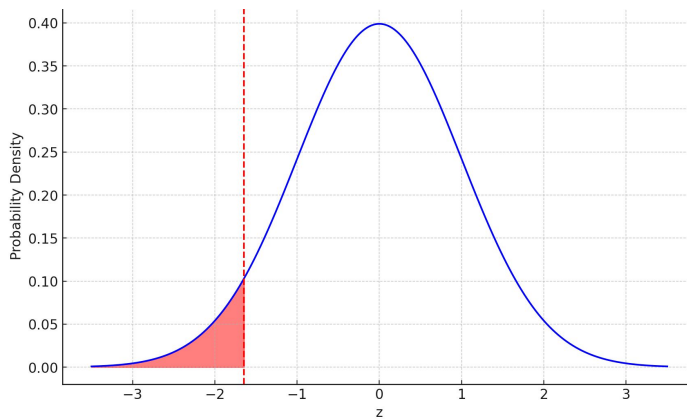




Right-tailed test



Left-tailed test



Two-tailed test

Values above
and below the mean

?

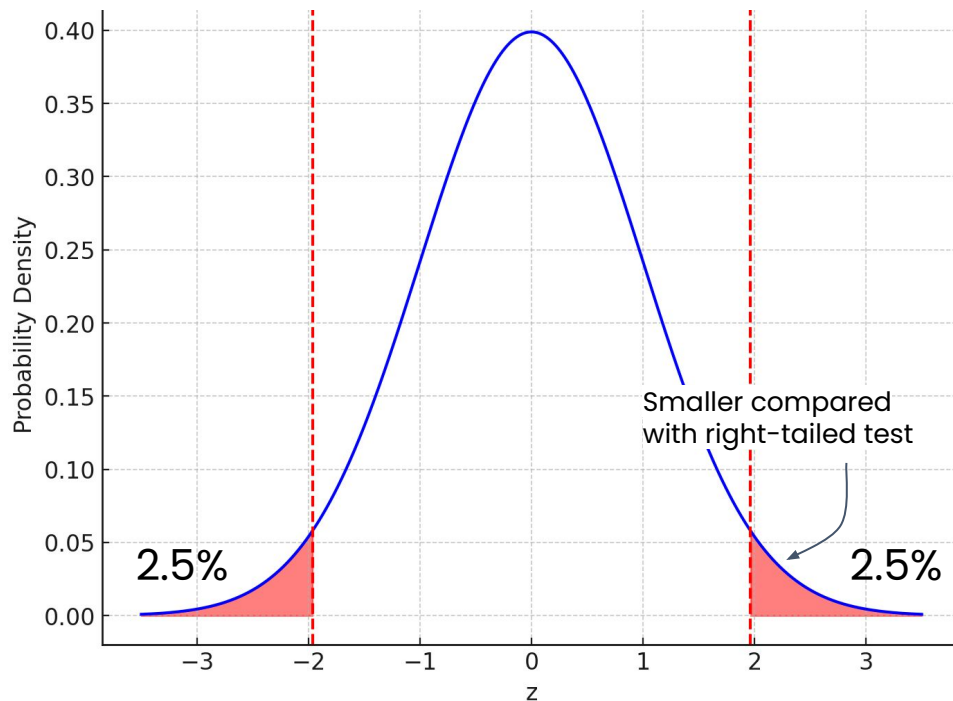


Two-tailed test

$$H_0 : \mu = 120 \text{ mins}$$

$$H_1 : \mu \neq 120 \text{ mins}$$

- Rejection region contains 2.5% of the data on either side
 - Maintains precision, with errors only 5% of the time
- ✗ If both contained 5% of the values → error rate of 10%



Terminology

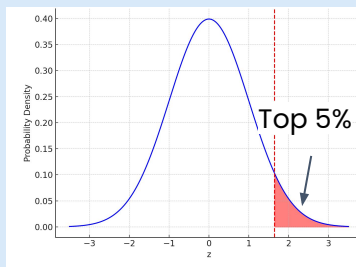
α

significance level

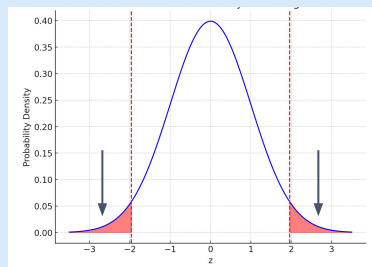
- Defines strength of evidence required to reject H_0
- Larger α lowers bar for strength of evidence
- Smaller α requires stronger evidence against H_0
- Common values for α : **0.10**, **0.05**, and **0.01**

Rejection region

- Area of sampling distribution that contains values that would lead you to reject the H_0



Movie lengths that are surprisingly long



Testing if pool pH is different from 7.4

Hypothesis testing

Calculating the p value

Rarity of the test statistic



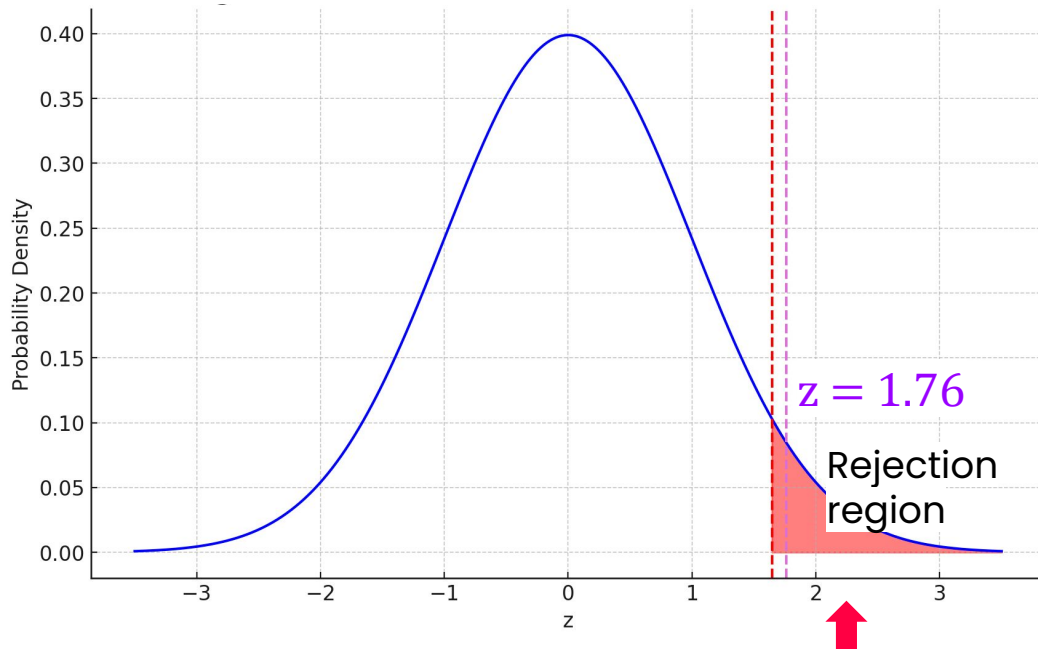
$$\alpha = 0.05$$

$$z = 1.76$$

How rare is this value?

$\bar{x} = 123$ minutes would happen
less than 5% of time

if $\mu = 120$ minutes





Calculating the p-value

$$\alpha = 0.05$$

$$z = 1.76$$

Calculate exactly how rare using:

p-value

- Probability of $\bar{x}$ as rare or rarer than z in direction of H_1
- Probability of $z = 1.76$ or higher

You'll need one of two things:

- ☐ Lookup table
- ☐ Spreadsheet or programming language

Calculating p-values with the CDF



$$\alpha = 0.05$$

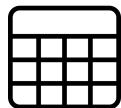
$$z = 1.76$$

Calculate: probability of $z = 1.76$ or higher

$$P(z > 1.76) = 1 - P(z \leq 1.76)$$

$$= 1 - 0.9608$$

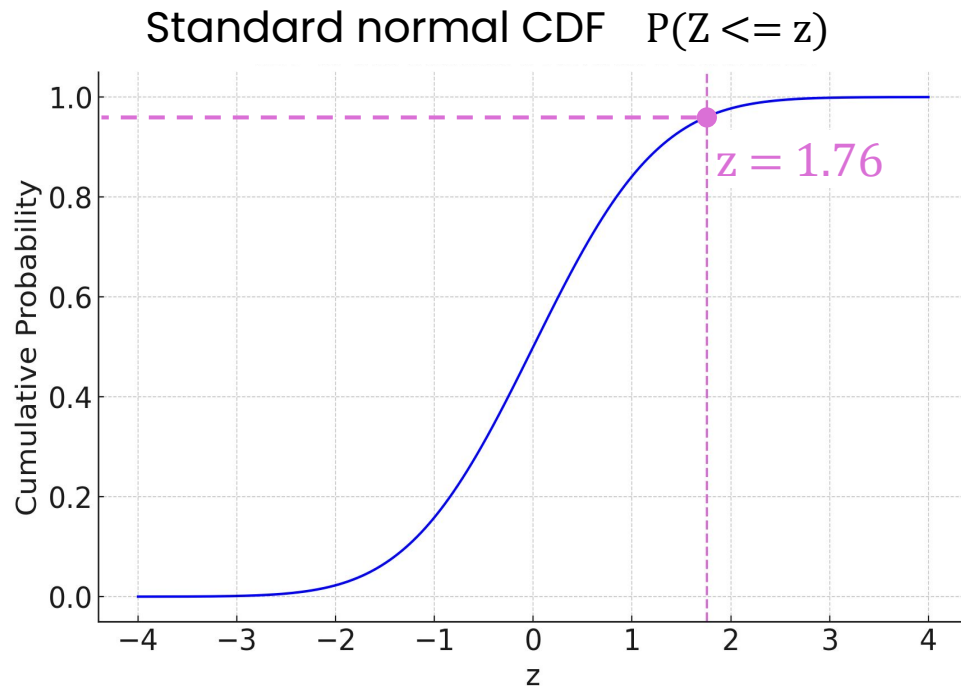
$$\text{p-value} = 0.0392$$



Sample data →



Z.TEST





Interpreting the p-value

$$\alpha = 0.05$$

$$z = 1.76$$

$$p\text{-value} = 0.0392$$

If $\mu = 120$,

you would observe $\bar{x} = 123$ minutes
or longer

about **3.92%** of the time.

Compare p-value with α :

$$0.0392 < 0.05$$

Since p-value is below α , you **reject** H_0 .

Is this event expected to happen **less than 5% of the time?**

Yes!



Interpreting the p-value

$$\alpha = 0.01$$

$$z = 1.76$$

$$\text{p-value} = 0.0392$$

If $\mu = 120$,

you would observe $\bar{x} = 123$ minutes
or longer

about **3.92%** of the time.

Compare p-value with α :

$$0.0392 > 0.01$$

You **fail to reject** H_0 , since p-value is above α .



Determine α before testing!

Scenario

Are movie durations longer than 120 mins?

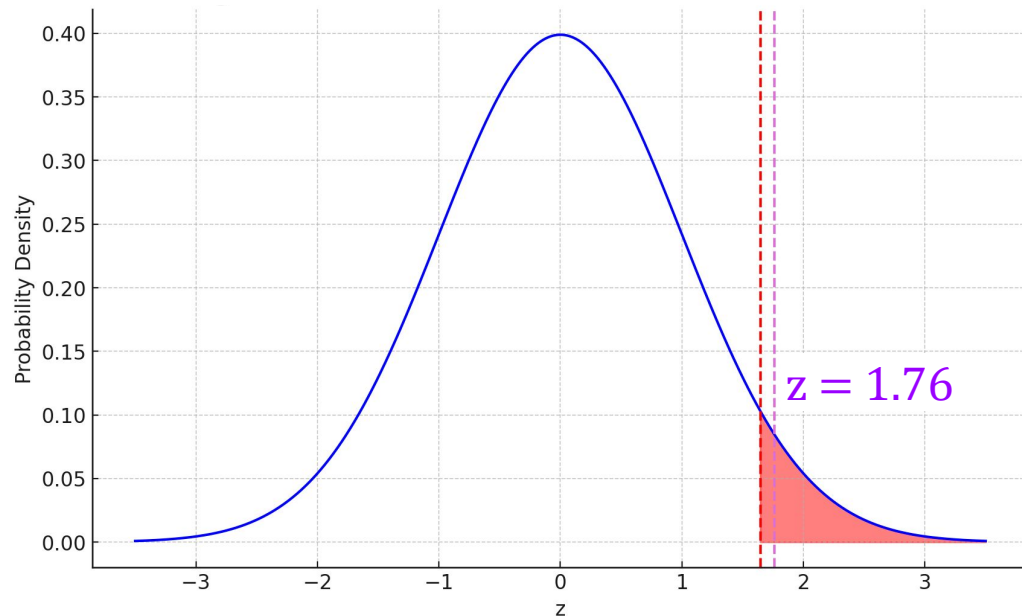
- $n = 50$ movies
- $\bar{x} = 123$ minutes
- $s = 12$ minutes

With 95% confidence in your conclusion:

- Rejected H_0
- Sufficient evidence that μ is > 120 mins

Chance of observing $\bar{x} \geq 123$:

p-value = 0.0392



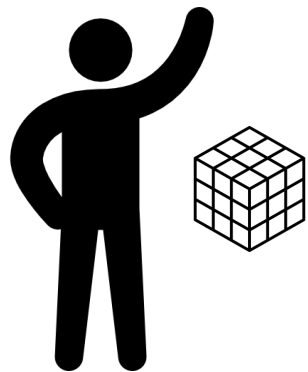
Hypothesis testing

Demo: hypothesis testing
for means

Hypothesis testing

Hypothesis testing errors

Scenario



You
Data Analyst



Problem: Who's the better solver?

Real life

Only two possible realities:

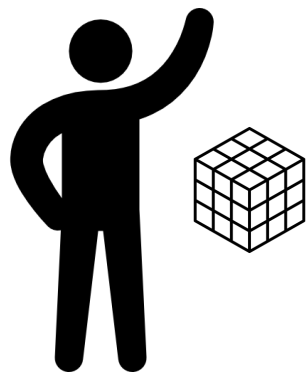
1. You're equally matched
2. One person is better

Hypothesis test

Two conclusions:

1. No significant difference between you
2. One person is significantly better

True effects



You
Data Analyst

True negative

No difference to find

Real life

Only two possible realities:

1. You're equally matched
2. One person is better



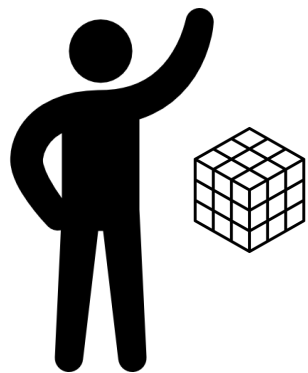
Didn't find significant difference

Hypothesis test

Two conclusions:

1. No significant difference between you
2. One person is significantly better

True effects



You
Data Analyst

True positive

Difference in the real world

Real life

Only two possible realities:

1. You're equally matched
2. One person is better

Test correctly identified it

Hypothesis test

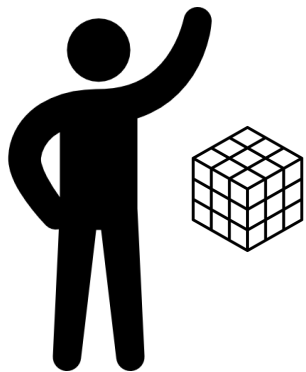
Two conclusions:

1. No significant difference between you
2. One person is significantly better



Errors

False positive



You
Data Analyst

No difference in reality

Real life

Only two possible realities:

1. You're equally matched
2. One person is better

Found a difference

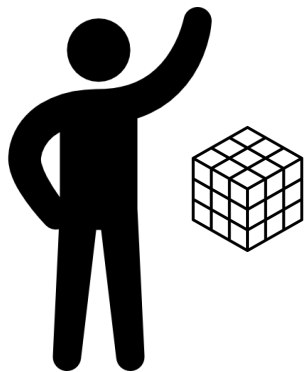
Hypothesis test

Two conclusions:

1. No significant difference between you
2. One person is significantly better

Errors

False negative



You
Data Analyst

Difference in reality

Real life

Only two possible realities:

1. You're equally matched
2. One person is better

Failed to find a difference

Hypothesis test

Two conclusions:

1. No significant difference between you
2. One person is significantly better



True effects and errors

True state of the world

Conclusion	H_0 is true	H_0 is false
Reject H_0	 False positive	 True positive
Fail to reject H_0	 True negative	 False negative

Controlling error rates

1 Significance level α

Chance of incorrectly rejecting the H_0

- ✓ Same as the false positive rate
- ✓ Can minimize false positives by setting α to a very low value

- ✗ Too low: increase false negatives
- ✗ Too low: very strict about what counts as significant
- ✗ Too low: harder to find true effects

2 Sample size

Increasing or planning to have large sample

- ✓ Provides a more accurate estimate
- ✓ Helps reduce both types of errors
- ✓ Helps detect very subtle effects

Choosing which error to minimize

Sector	False positive	False negative	Preferred error to minimize
 Manufacturing	Identifying product as defective when it is fine	Not detecting defective product	False positives if customers can return
 Medical testing	Over-diagnose patient	Patient has illness, but test fails to detect it	False negatives to ensure lifesaving treatment
 Loan approval	Approving loan for someone who defaults	Denying loan to someone who would pay back	Balance between both



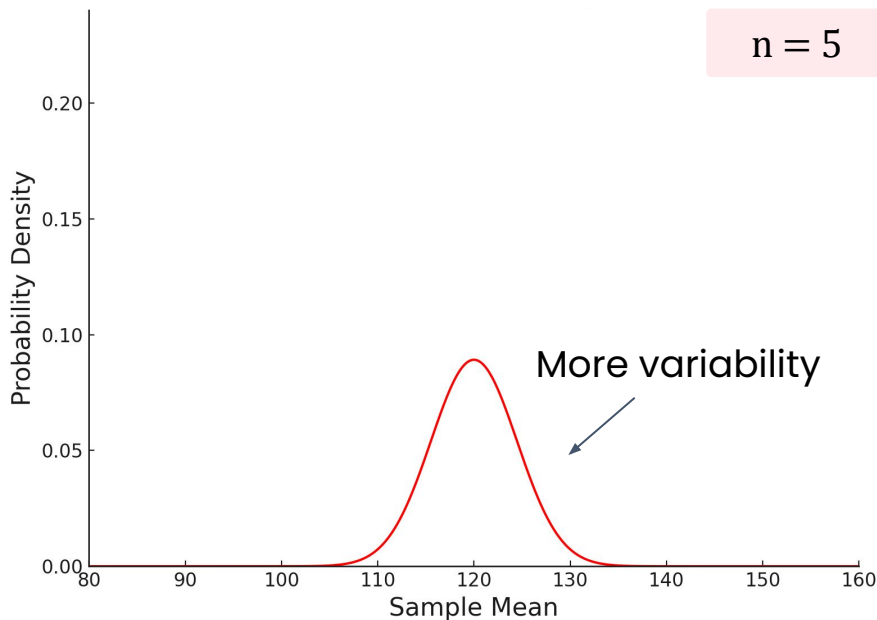
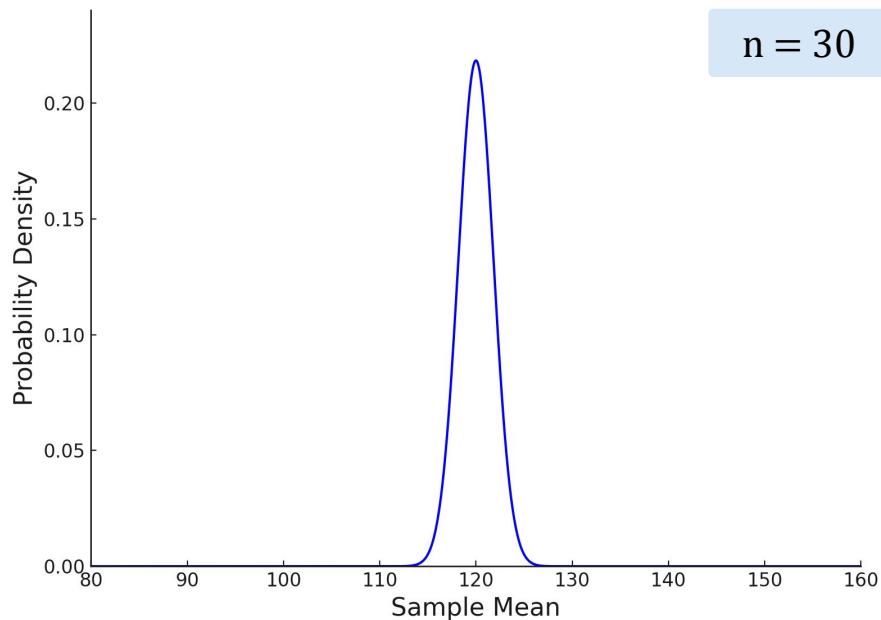
Hypothesis testing

The t distribution

$$SE = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}} - \text{Introduce more uncertainty}$$

$n < 30$ is common
 σ is rarely known

Use t distribution



t distribution

The normal and t distributions are similar:

- Smooth shape
- Symmetric about the mean of 0

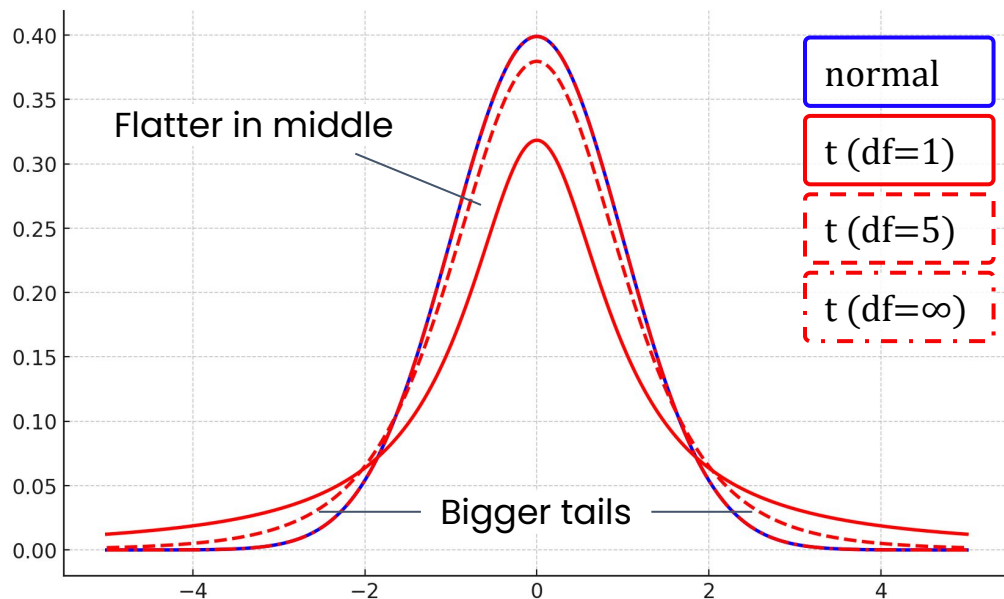
The t distribution is:

- Flatter in the middle
- Has bigger tails
- Defined by **degrees of freedom** (df)

$$df = n - 1$$

← Using s instead of σ
to calculate SE

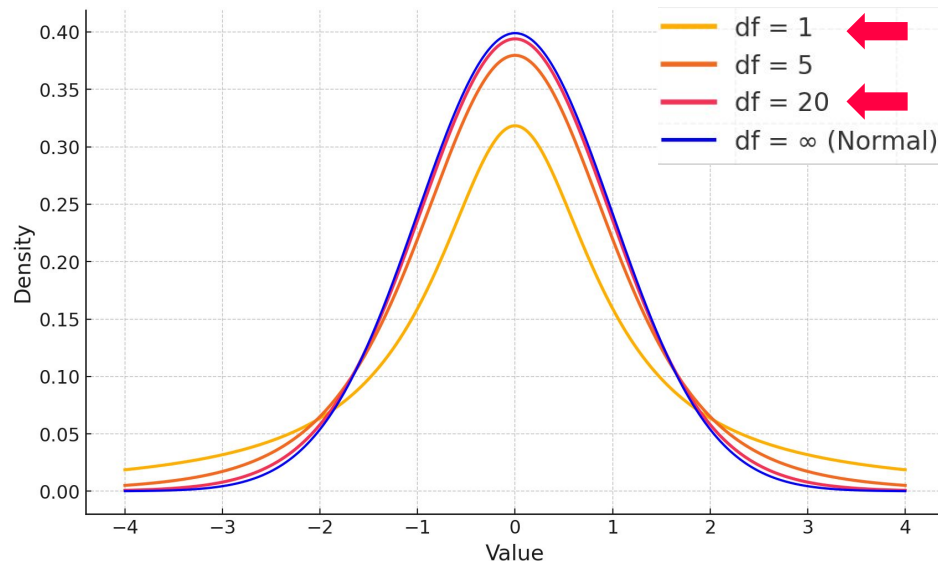
- As df increases, t distribution is more similar to normal distribution
- Choice of distribution becomes less important
- As n increases, you get better information about variability of population



t distributions

The fundamental idea is:

- In order to calculate a confidence interval or perform a hypothesis test, you generally don't know the population standard deviation
- As your sample size increases, you get better information about variability of the population
- With smaller samples, you're working with a less precise estimate



Hypothesis tests with the t distribution

	Normal distribution	t distribution
Defining hypotheses	Same!	
Calculate sample statistics	Same!	
Calculate test statistic	Normal distribution (z)	t-distribution (t)
Spreadsheet function	z . test	t . test

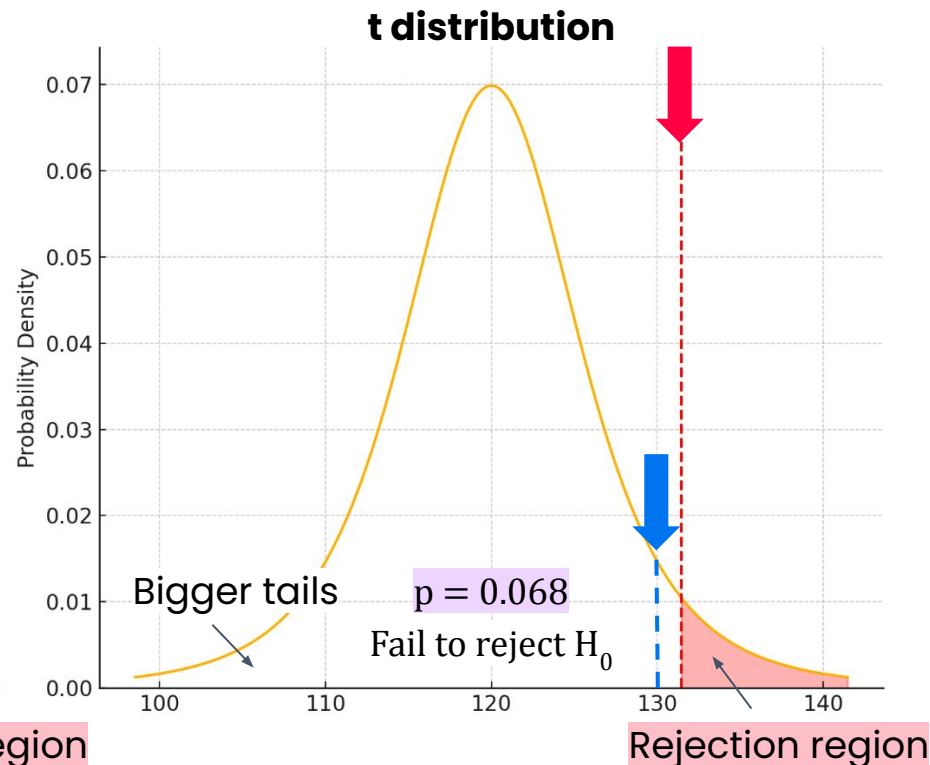
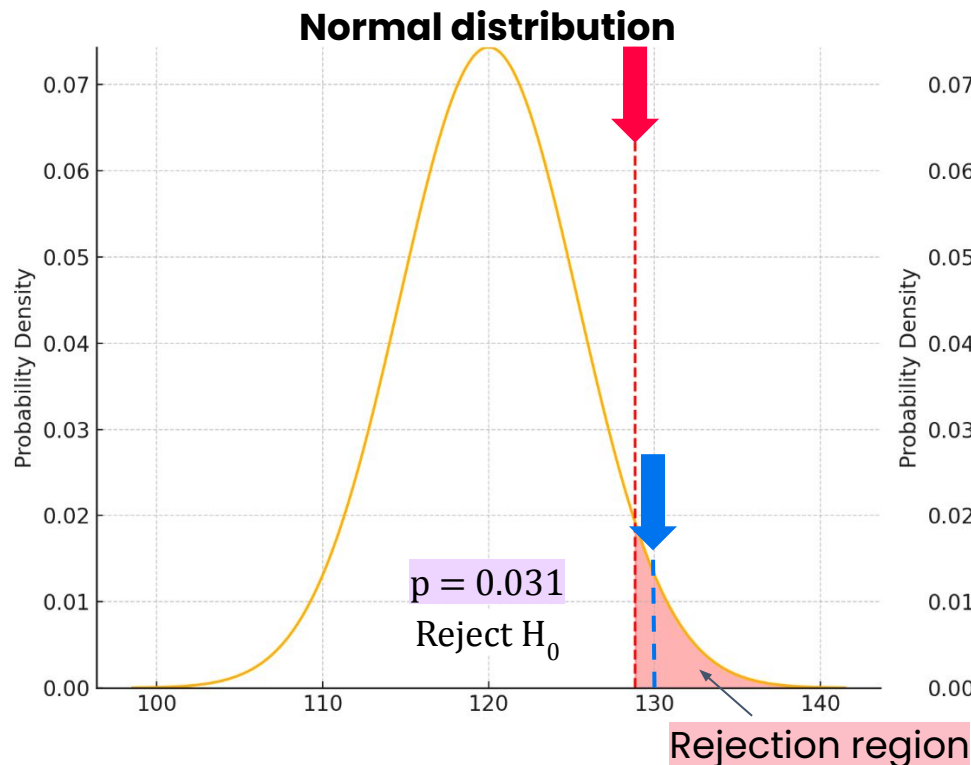
Hypothesis testing

$\bar{x} = 130$ mins

$\mu = 120$

$s = 12$

$n = 5$



Application of the t distribution

t distribution

$$n < 30$$

★ Still relevant!



Researching rare illnesses



Studying endangered species

Normal distribution

$$n \geq 30$$



Big data



Tech



Manufacturing



Healthcare



Hypothesis testing

Hypothesis testing
for proportions

Proportions



What proportion of coworkers is in favor of birthday card?



Is proportion of valid canine DNA test kits 0.7?



What proportion of deliveries to zoo are on time?

Scenario



You
Data Analyst

Problem:

Whether true proportion of late deliveries is **greater than 5%**



More than 5% of
deliveries are late



Risk jeopardizing
contract with the zoo

You would want to:



Move up delivery driver's start time



Hypothesis testing for proportions

- $n = 250$ deliveries



→ $\hat{p}$ — Estimate for true proportion p

Hypothesis testing

- Define H_0 and H_1
- Determine α
- Calculate z and p -value
- Interpret results and make a decision

$$H_0 : p = 0.05$$

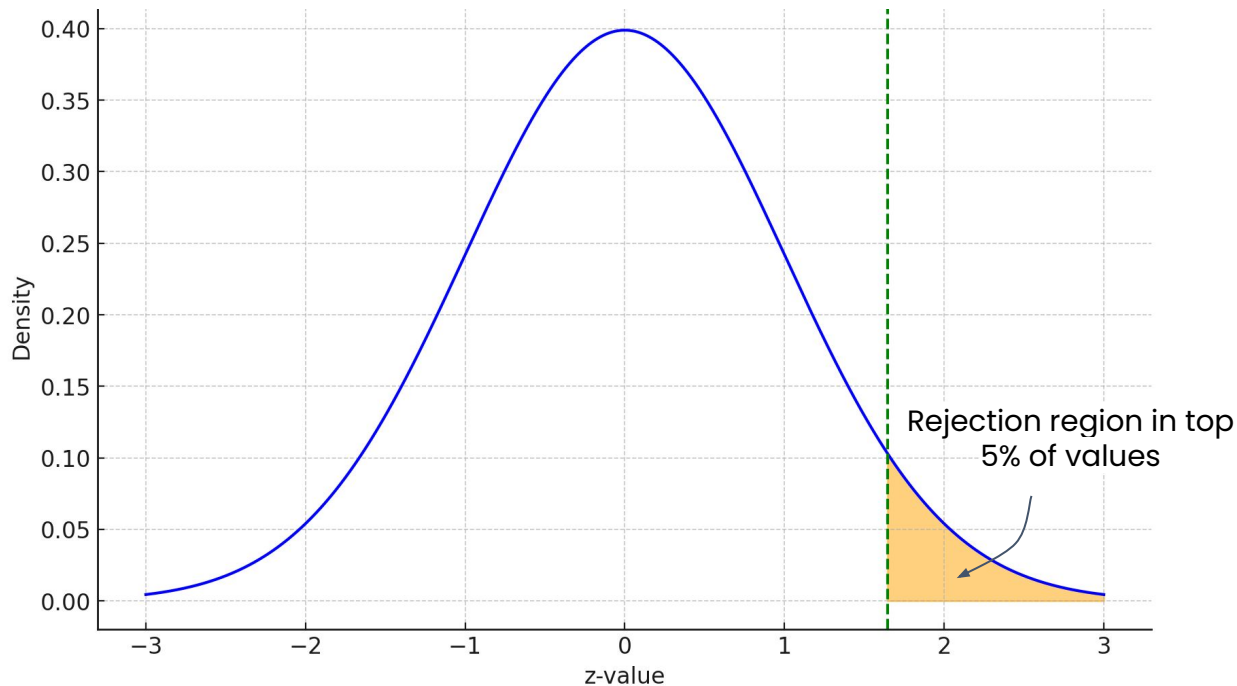
$$H_1 : p > 0.05$$

$$\alpha = 0.05$$

→ 5% false positive rate



- $n = 250$



$$H_0 : p = 0.05$$

$$H_1 : p > 0.05$$

$$\alpha = 0.05$$



Calculate test statistic

- 1 Measured sample proportion ($\hat{p}$) of late deliveries

 **Late** $\rightarrow \hat{p} = 0.06$

- 2 Calculate test statistic z

1. Subtract hypothesized proportion from $\hat{p}$
2. Divide by the SE for the distribution

$$SE \rightarrow \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Centers z at 0

How many SEs $\hat{p}$ is from p_0

Why not use $\hat{p}$ for standard error?

Means

- μ is hypothesized
- σ often unknown
- Use s to approximate

Proportions

- Derive σ from p_0 and n
- Don't need to estimate



Calculate test statistic

- 1 Measured sample proportion ($\hat{p}$) of late deliveries

 **Late** $\rightarrow \hat{p} = 0.06$

- 2 Calculate test statistic z

1. Subtract hypothesized proportion from $\hat{p}$
2. Divide by the SE for the distribution

SE $\rightarrow \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$

$\hat{p} - p_0$ $\leftarrow$ Centers z at 0

$\sqrt{\frac{p_0(1-p_0)}{n}}$ $\leftarrow$ How many SEs $\hat{p}$ is from p_0

$$\frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.06 - 0.05}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.01}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.01}{\sqrt{\frac{0.05 \times 0.95}{n}}} = \frac{0.01}{\sqrt{\frac{0.0475}{n}}} = \frac{0.01}{\sqrt{\frac{0.0475}{250}}} = \frac{0.01}{\sqrt{0.00019}} = \frac{0.01}{0.01378} = 0.725$$

z is above the mean by less than 1 SE

Calculate test statistic

How rare would a value of 0.725 or higher be?

$$z = 0.0725$$

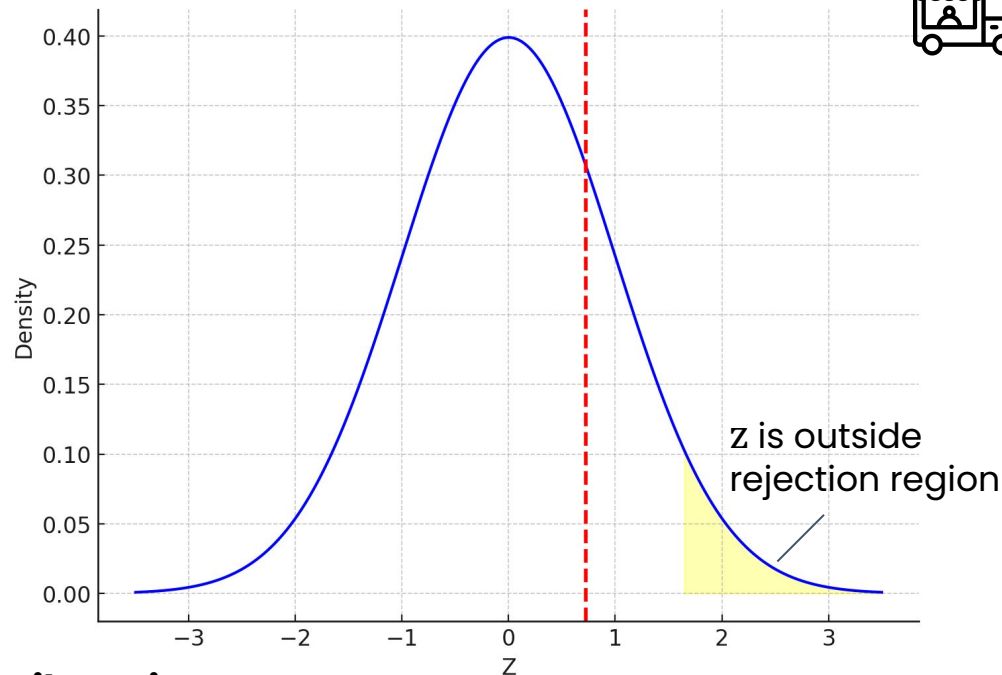
Do you think you would reject or fail to reject H_0 ?

Fail to reject

$$H_0 : p = 0.05$$

$$H_1 : p > 0.05$$

$$\alpha = 0.05$$



Calculate p-value and compare with α



3 Calculate p-value

You can use:

- ☐ A lookup table
- ☐ Software

$$P(z \geq 0.725) = 0.2342$$

If $p = 0.05$, 23.42% of $\hat{p}$ s will be ≥ 0.05

4 Compare p-value with α

$$\alpha = 0.05$$

- 0.373 is greater than α
- Fail to reject H_0

$$H_0 : p = 0.05$$



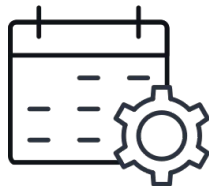
Scenario



You
Data Analyst

Your interpretation of the results:

- **Not sufficient** evidence to conclude that late deliveries happen $> 5\%$ of the time.



Keep start time the way it is



Contract with zoo should remain in good standing

Hypothesis testing

Demo: hypothesis testing
for proportions



Hypothesis testing

Two sample tests

Two sample tests



Do movies from 2013 have the same duration as movies from 1934?



Is proportion of people with positive opinion of AI the same across age groups?



Do users with basic and premium stay subscribed for different lengths?



Two sample tests

Do users with **basic** and **premium** stay subscribed for **different** lengths?

$$\alpha = 0.05$$

Hypothesis testing

1. Define H_0 and H_1
2. Determine α
3. Calculate z or t and p-value
4. Interpret results and make a decision

- $H_0 : \mu_{\text{basic}} = \mu_{\text{premium}} \rightarrow \text{no difference}$
- $H_1 : \mu_{\text{basic}} \neq \mu_{\text{premium}} \rightarrow \text{means are different}$



Two sample tests

Do users with **basic** and **premium** stay subscribed for **different** lengths?

$$\alpha = 0.05$$

Next step: calculate test statistic (t)

- ☐ Spreadsheet
- ☐ Programming language

Basic subscribers:

$$n_{\text{basic}} = 30$$

$$\bar{x}_{\text{basic}} = 9.9 \text{ months}$$

$$s_{\text{basic}} = 3.3 \text{ months}$$

Premium subscribers:

$$n_{\text{premium}} = 30$$

$$\bar{x}_{\text{premium}} = 10.4 \text{ months}$$

$$s_{\text{premium}} = 1.9 \text{ months}$$

Independent

Two sample tests



Do users with **basic** and **premium** stay subscribed for **different** lengths?

$$\alpha = 0.05$$

Next step: calculate test statistic (t)

- ☐ Spreadsheet
- ☐ Programming language

Basic subscribers:

$$n_{\text{basic}} = 30$$

$$\bar{x}_{\text{basic}} = 9.9 \text{ months}$$

$$s_{\text{basic}} = 3.3 \text{ months}$$

Premium subscribers:

$$n_{\text{premium}} = 30$$

$$\bar{x}_{\text{premium}} = 10.4 \text{ months}$$

$$s_{\text{premium}} = 1.9 \text{ months}$$

Independent
Subscription for **basic**
not influenced by **premium**



Promotional rate



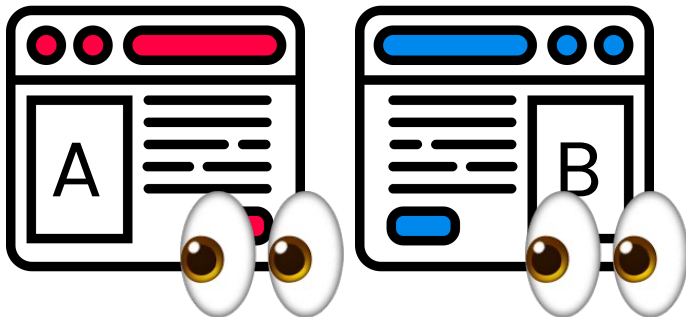
Basic



Premium

Two sample tests

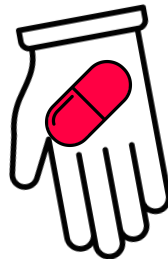
A/B testing



Do groups respond differently?

Clinical research

Experimental



Control



Compare whether experimental group showed more benefits

Hypothesis testing

Other hypothesis tests

Keep in mind...



Don't need to memorize each of the tests



As you need them, you can look up specifics



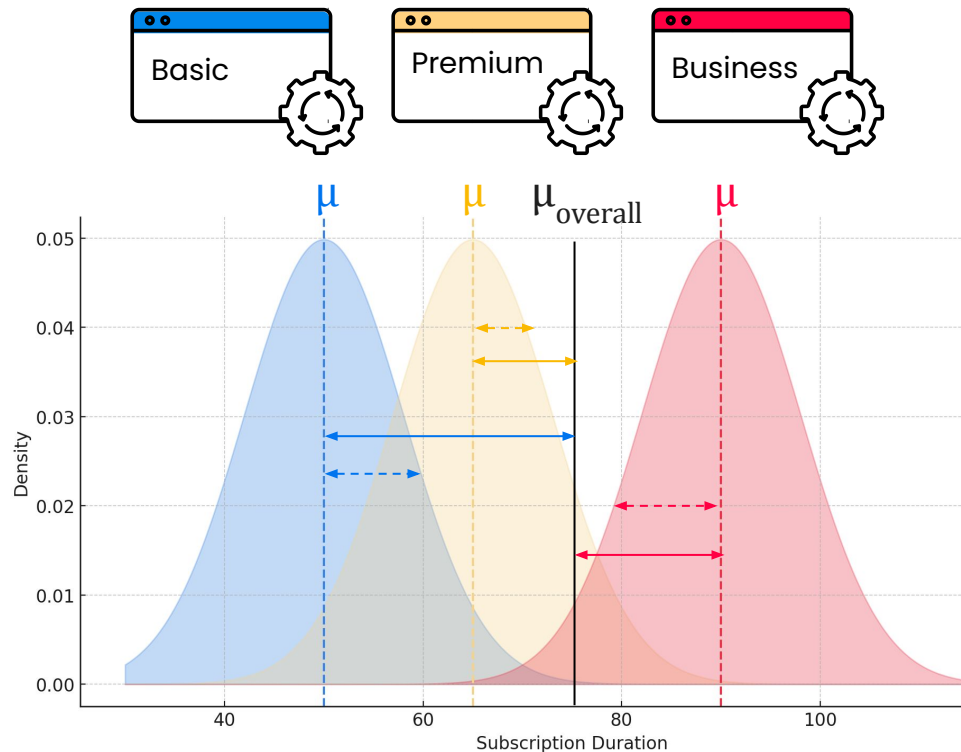
You already have foundational knowledge to conduct and interpret them

ANOVA

Scenario: Comparing three or more groups

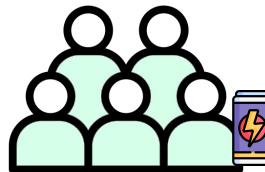
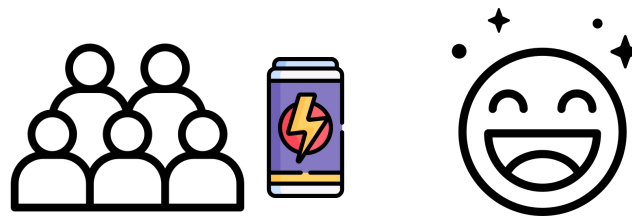
Test: Analysis of variance (ANOVA)

1. Calculate the means for each group
2. Calculate the overall mean
3. Compare:
 - How much group means differ from overall mean
 - How much scores differ from group means
4. **A small p-value (typically < 0.05):** suggests differences between groups are significant

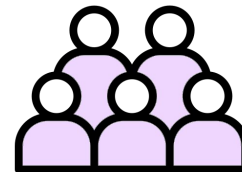


Paired t-test

Scenario: Working with data that represents a before and after condition



Random sample 1



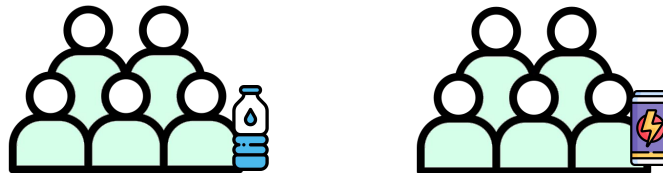
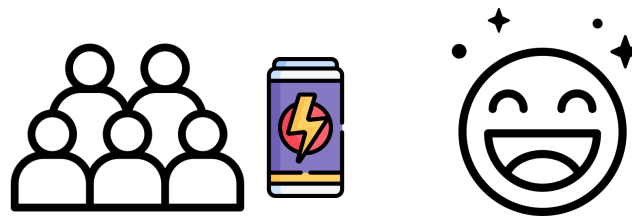
Random sample 2

Paired t-test

Scenario: Working with data that represents a before and after condition

Test:

1. Test the same group of people twice
2. Calculate difference between each pair of measurements
3. Calculate the test statistic by dividing the mean difference by SE
4. **p-value (< 0.05)**: often considered significant



More information!

Chi-squared test

Scenario: Working with categorical data

Test:

1. Create table of observed frequencies
2. Calculate expected frequencies assuming no relationship
3. Calculate chi-squared statistic: how much observed frequencies deviate



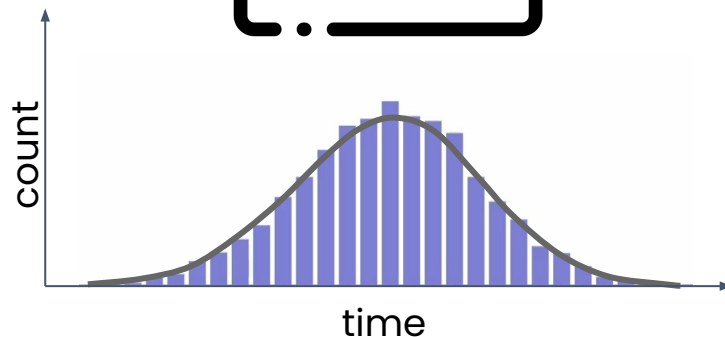
Satisfaction ratings

Goodness of fit test

Scenario: Test whether data follows a particular distribution

Test:

1. Calculate test statistic measuring deviation from distribution
2. Determine p-value



Hypothesis testing

Interpretation with LLMs

Hypothesis testing

Inference with LLMs



Applied Statistics for Data Analytics

Your next steps